Survey of the Human Proteostasis Network: The Autophagy-Lysosome Pathway

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Abstract

The condition of having a healthy, functional proteome is known as protein homeostasis, or proteostasis. Establishing and maintaining proteostasis is the province of the proteostasis network, a highly interconnected system of over 3000 components that regulate protein synthesis, folding, localization, and degradation. The proteostasis network is a fundamental functional unit in biology that is essential for cellular health and has direct relevance to many diseases of protein conformation. However, it is not well defined or annotated, hindering its functional characterization in health and disease. In this series of manuscripts, we aim to operationally define the human proteostasis network by providing a comprehensive, annotated list of its components. In the other manuscripts in this series, we survey the components of the proteostasis network that make up (1) the machineries for protein synthesis, protein folding, protein trafficking into and out of organelles, and organelle-specific degradation pathways and (2) the ubiquitin-proteasome system. Here, we have compiled a curated list of over 850 unique high-confidence components of the autophagy-lysosome pathway, the other major protein degradation system in human cells. This resource will facilitate the rigorous investigation of autophagy-lysosome dysfunction as a driver of proteostasis imbalance in aging and disease.

Keywords

Annotation; degradation; neurodegenerative disease; aging; systems; lysosomal catabolism

Abbreviations

ALP: Autophagy–lysosome pathway; AMPK: AMP-activated protein kinase; ATG: Autophagy-related (used as a prefix for autophagy-related genes and proteins, e.g., ATG4A, ATG8); BORC: BLOC-one-related complex; CMA: Chaperone-mediated autophagy; ER: Endoplasmic reticulum; ESCRT: Endosomal complexes required for transport; GABARAP: GABA type A receptor-associated protein; HOPS: Homotypic fusion and protein sorting (complex); LAMP1 / LAMP2: Lysosome-associated membrane protein 1 / 2; LIR: LC3-interacting region; MAP1LC3: Microtubule-associated protein 1 light chain 3; mTORC1: Mammalian target of rapamycin complex 1; PI: Phosphatidylinositol; PI(3)P: Phosphatidylinositol 3-phosphate; PI3KC3: Class III phosphatidylinositol 3-kinase; PN: Proteostasis network; SNARE: Soluble N-ethylmaleimide–sensitive factor attachment protein receptor; ULK: UNC-51-like autophagy

activating kinase; UPS: Ubiquitin–proteasome system; WIPI: WD repeat domain, phosphoinositide interacting

Introduction

Proteostasis is the maintenance of the proteome in a healthy, functional state [1]. Proteostasis is enabled by the proteostasis network, a collection of cellular components responsible for managing the synthesis, folding, trafficking, and degradation of proteins [1-5]. Although the term "proteostasis network" was introduced over a decade ago [1], the network has remained poorly defined as there is no comprehensive accounting of its components. This problem is especially acute considering the important role of proteostasis in aging-related neurodegenerative diseases [6-11]. To address this problem, we have been creating a detailed enumeration of the proteostasis network. Here we report a comprehensive list of the components of the autophagy-lysosome pathway (ALP). In two other manuscripts in this series we report surveys of (1) the machinery of protein synthesis, chaperones, folding enzymes, the systems for trafficking proteins into and out of organelles, and organelle-specific degradation pathways and (2) the ubiquitin-proteasome system (UPS).

The molecular architecture of the ALP is complex, encompassing core autophagy machinery, regulatory signaling modules, vesicle trafficking factors, lysosomal components, and selective autophagy receptors. These elements operate across multiple cellular compartments and engage in extensive crosstalk with other arms of the proteostasis network [12-21]. Unlike the ubiquitin–proteasome system, which primarily degrades short-lived and soluble proteins, the autophagy–lysosome pathway is uniquely capable of eliminating large protein assemblies, protein aggregates, and organelles [22,23]. Through distinct but interconnected processes—including bulk and selective macroautophagy, microautophagy, and chaperone-mediated autophagy—the ALP enables cells to remodel their proteome and organelle content in response to stress, nutrient availability, etc., thus occupying a central position at the interface of protein quality control, organelle homeostasis, and cellular adaptation [12-21].

Within the proteostasis network, the ALP is of particular interest because mutations in many of its components have been found to increase the risk of diseases of failed proteostasis, like Alzheimer disease, frontotemporal dementia, Parkinson disease, and amyotrophic lateral sclerosis, among others [11,24,25]. The most prominent diseases associated with ALP dysfunction are neurodegenerative. This observation illustrates the importance of autophagy for maintaining proteostasis in post-mitotic neurons. For example, failures in the ALP produce striking

pathological signatures in the lysosomes of Alzheimer's disease models and patients [26]. However, it is important to note that dysregulated autophagy also has a role in cardiovascular, musculoskeletal, bone, pulmonary, kidney, and liver diseases, as well as metabolic syndromes and cancer [27].

In this manuscript, we present a description of the ALP in terms of a five-tiered taxonomic annotation system. This enables us to fit the ALP into a consistent framework with the rest of the proteostasis network, including the translation, folding, trafficking and regulatory pathways as well as the ubiquitin-proteasome system. In doing so, we have relied on previous efforts to enumerate the components of the ALP [28-32] as a foundation upon which we have built our updated, expanded, and systematic annotation. Through this process, we define a total of over 850 unique ALP components. We anticipate that this harmonized taxonomy of the ALP, explicitly aligned with the broader architecture of the proteostasis network, will serve as a valuable resource for the field and will facilitate a deeper and more integrated understanding of how the ALP contributes to organismal health and disease within the context of global proteostasis.

Results

Taxonomic system for annotating the ALP

Our taxonomic annotation system for the proteostasis network was constructed to convey at a glance a component's role in proteostasis. It consists of five levels: Branch, Class, Group, Type, and Subtype. The Branch annotation is the broadest and refers to a component's localization or membership in an overarching pathway. There are nine Branches of the proteostasis network (PN): cytonuclear proteostasis, ER proteostasis, mitochondrial proteostasis, nuclear proteostasis, PN regulation, translation, extracellular proteostasis, the ALP, and the UPS (Figure 1). As noted above, the ALP is the focus of this manuscript. The next level of the taxonomy scheme is Class, which refers to a component's function in proteostasis (e.g., autophagophore initiation and elongation, autophagy substrate selection, etc.). The Classes associated with each Branch are shown in Figure 2. The Group, Type and Subtype annotations give increasingly specific descriptions of a component's role in proteostasis. We endeavored to use as few descriptors as possible for each component. Thus, many components lack Type and Subtype annotations. Also,

some components have multiple roles in proteostasis and therefore have multiple sets of annotations.

For some Branches of the PN, structure and function are closely tied together and categorizing components based on structural features is an effective way to organize their annotations. Examples include the chaperone-based Classes in the cytosolic proteostasis Branch and the E3 ligases in the UPS Branch. In contrast, this method of systematizing is not effective in the ALP. Many different components, usually with an array of different structures, are brought to bear in a given ALP process. Thus, we have organized the ALP Branch around the major stages of autophagy. We first provide a brief overview of the pathway to orient the reader (Figure 3). Autophagy is governed by conserved signaling pathways that regulate flux through the autophagy–lysosome pathway (ALP) [33]. These inputs—including, for example, nutrient, hormone, energy, and stress signals—are mostly communicated through mTORC1 (mammalian target of rapamycin complex 1) and AMPK (AMP-activated protein kinase), which are regulators of numerous key pathways in addition to autophagy, like cell metabolism and energy homeostasis [34-39]. These kinases modulate autophagy through modification of the ULK complex (unc-51 like autophagy activating kinase), with mTORC1 and AMPK exerting inhibitory and activating effects, respectively, through phosphorylation at specific sites, (for example, Ser757 of ULK1 by mTORC1 and Ser317/Ser777 of ULK1 by AMPK) [35,38]. When the ULK complex is activated, it is recruited to the site where the autophagophore (as the nascent autophagosome is known) formation is initiated; sometimes, this is at a structure in the ER membrane known as the omegasome [40]. There, the ULK complex activates the PI3KC3 complex 1 (class III phosphatidylinositol 3-kinase)[41] which phosphorylates phosphatidylinositol (PI) at the 3 position to yield phosphatidylinositol 3-phosphate (PI(3)P) [13,42-44]. PI(3)P recruits the ATG2 (autophagy related 2)-WIPI (WD repeat domain, phosphoinositide interacting) complex which rapidly transports lipids to the growing autophagophore [13,45,46]. The ATG2-WIPI complex also recruits the ATG5-ATG12-ATG16 complex [47], which catalyzes the conjugation of phosphatidylethanolamine to human ATG8 orthologs like the MAP1LC3 (microtubule associated protein light chain 3) or GABARAP (GABA type A receptor-associated protein) proteins on the autophagophore membrane [48-50]. These lipidated proteins are critical components of the autophagosome [51-55], especially for substrate recognition [56-59].

The autophagophore grows until it surrounds its substrate, whether that is a portion of cytosol, as in bulk macroautophagy, or other substrates such as protein aggregates or entire organelles, as in selective autophagy [20]. The substrates of selective autophagy are bound by receptors that recognize substrates by various mechanisms (often involving ubiquitin [57]), and recognize autophagosomes via their displayed ATG8 homologs [56,58,59]. The autophagophore is sealed with the assistance of ESCRT complexes (endosomal complexes required for transport), yielding an autophagosome [13,60]. Autophagosomes are transported through the cell [18,61] to fuse with lysosomes, creating autolysosomes [14], in which the cargo from the autophagosome is degraded by lysosomal enzymes [62]. Lysosomes are then regenerated from the autolysosomes by the process of autophagic lysosome reformation [63].

Criteria for inclusion in the proteostasis network

We use two criteria to decide whether components should be included in the proteostasis network. The first is the "entity-based" criterion, in which entities (proteins, protein complexes, non-coding RNA, etc.) are included because there is experimental evidence for their having a role in proteostasis. The second is the "domain-based" criterion, in which components are included because they have at least one structural domain that is characteristic of protein families known to be involved in proteostasis. We primarily used the entity-based criterion applied to existing lists of autophagy components (for example, autophagy-related Gene Ontology annotations [28,64], the KEGG pathway "Autophagy-Animal" [65], and previous compilations of ALP components [29-31], especially the recent list by Bordi and co-workers [32], to populate our initial list of ALP components. This initial list was then expanded through a comprehensive literature search as described in the Methods section. The domain-based criterion was also used in a few cases. For example, MAP1LC3B2 was included because it is a homolog of the yeast protein ATG8 [54,66]. ATG8 is central to the yeast ALP [67,68] and six of its seven human homologs have also been shown to function in the ALP, but to our knowledge MAP1LC3B2 has not [54]. Nevertheless, we include it in our annotation by virtue of its homology to ATG8.

While the scope of the proteostasis network naturally encompasses the fate of proteins, our enumeration extends slightly beyond the bounds of the protein world. For example, our description of selective autophagy includes all forms of it, including those by which biomolecules other than

proteins are degraded, such as lipophagy [69]. We also included all the catabolic enzymes of the lysosome in the ALP, not just those that catalyze the cleavage of peptide bonds, as well as those components involved in maintaining the acidic lysosomal environment. However, we excluded components that are generally involved in lysosomal or endosomal trafficking and components associated only with non-canonical autophagy or with secretory autophagy [70-72]. These pathways may be included in later versions of the ALP when their significance and mechanisms are better understood.

Taxonomic Classes in the ALP are organized according to temporal events during autophagy

As noted above, the Class-level annotations in the ALP largely reflect the stages described above in the autophagy process and are illustrated in the schematic overview of the ALP in Figure 3. These Classes are "pre-initiation autophagy signaling"; "autophagophore initiation and elongation"; "autophagy substrate selection"; "autophagosome closure, maturation, and lysosome fusion"; "lysosomal catabolism"; and "autophagic lysosome reformation". The remaining Classes are "autophagy gene expression", which comprises the transcription factors and transcriptional and translational regulators that control the expression of autophagy genes [73]; "microautophagy", which includes the components required for the various forms of microautophagy, such as micromitophage, microreticulphagy, etc. [74]; "chaperone-mediated autophagy", which includes components involved in chaperone-mediated autophagy [75]; and "specific function in autophagy unknown", a small Class that contains components that clearly affect autophagy but through mechanisms that are not yet known. The Groups, Types, and Subtypes further specify a component's function within the processes denoted by their Class.

The application of our taxonomic scheme to an ALP component is illustrated in Figure 4 for ATG4A, a cysteine protease that trims the C-terminal Arg from ATG8 homologs in preparation for their conjugation to phosphatidylethanolamine [76,77]. The following annotations were assigned to ATG4A: (1) Branch: autophagy-lysosome pathway; (ii) Class: autophagophore initiation and elongation; (iii) Group: ATG8 homolog processing, direct; (iv) Type: preparation of ATG8 homologs for lipidation; (v) Subtype: protease that removes C-terminal Arg from ATG8. It is important to note that these annotations are not unique, as they are shared with ATG4A's paralogs in the human genome: ATG4B, ATG4C, and ATG4D [77]. Also, like many other ALP

components, ATG4A has two distinct sets of annotations because in addition to its role in preparing ATG8 homologs for lipidation, it is also responsible for delipidating ATG8 homologs and removing them from the autophagosome membrane as the autophagosome matures [76,78]. The annotation for this role of ATG4A in the ALP is: (i) Branch: autophagy-lysosome pathway; (ii) Class: autophagosome closure, maturation, and lysosome fusion; (iii) Group: regulation of autophagosome membrane composition; (iv) Type: ATG8 homolog de-lipidation; (v) Subtype: releases ATG8 homologs from maturing autophagosome.

Distribution of components in the ALP Branch

The complete list of components of the ALP is presented in Table S1 (see the Methods section for details on how the list was constructed) along with the rest of the proteostasis network (see the Data Availability Statement at the end of the main text for a link to download Table S1). There are 1048 entries in our list of ALP components, representing 856 unique components. All the entries have a Class annotation, of which there are ten in the ALP Branch, and all but four have one of the 81 Group annotations. The four exceptions are those components in the "specific function in autophagy unknown" Class. Most of the entries have one of the 150 Type annotations (966 out of 1048), but fewer than half of the entries have one of the 109 Subtype annotations (435 out of 1048). To illustrate how the finer-grained annotations were assigned in the ALP, we show how the components of the Class "autophagosome closure, maturation, and lysosome fusion" are distributed within this Class's various Groups, Types, and Subtypes in Table 1.

As noted above for ATG4A, many components of the ALP have multiple roles in autophagy. Of the 856 components in the ALP, 123 are listed in two or more Classes, indicating that they function at substantially different points along the autophagy pathway. These Class-level overlaps are illustrated in Figure 5. The most common types of components that have multiple Class-level functions are the ATG8 homologs (like MAP1LC3B, GABARAP, etc.) and the LIR-domain-containing components that recognize ATG8 homologs (like SQSTM1, CALCOCO2, etc.). This observation highlights that while these ALP components share responsibility for substrate delivery to lysosomes (thereby placing them in the autophagy substrate selection Class), their ability to recognize each other enables them to serve in other capacities as well, such as modulation of autophagophore growth (which places some of them in the autophagophore

initiation and elongation Class) or transport of the autophagosome (which places some of them in the autophagosome closure, maturation, and lysosome fusion Class). Overlap between the autophagophore initiation and elongation and autophagosome closure, maturation, and lysosome fusion Classes is also common for a variety of reasons. For example, we mentioned above that ATG4A/B/C/D cleave the C-terminal residue from ATG8 homologs to prepare them for lipidation and cleave the C-terminal lipid from lipidated ATG8 homologs to remove them from autophagosomes. The PI3KC3 complexes 1 and 2 offer another example of such dual use. These complexes only differ by one component (ATG14L vs. UVRAG) but the role of complex 1 is mainly in autophagy initiation while that of complex 2 is in autophagosome maturation and recruitment of the HOPS complex for autophagosome-lysosome fusion [12,79]. Thus, the functionality of the PI3KC3 complex is recycled and redirected to different tasks depending on its subunit composition. We speculate that the use of the same components at different points in the ALP in these and many other cases likely enforces coordination of the fluxes through the various stages of autophagy, by creating situations in which the same components are rate-limiting for different steps.

The ALP is integrated with other Branches of the proteostasis network

In addition to the overlaps described above in which different Classes of the ALP Branch share components, the ALP shares 223 out of its 856 components with the other Branches of the proteostasis network (Figure 6). The most extensive overlap is with the UPS; the ALP shares 156 components with the UPS (130 of these are shared with only the UPS; the remaining 26 are shared with at least one other Branch). This circumstance is in part because the ATG8 lipidation system is related to the ubiquitination system used by the UPS [80], and is therefore listed in both Branches. More importantly, however, it is because ubiquitination marks substrates for degradation by both the ALP and the UPS [22,23]. This enables each of these central degradative pathways to serve as a backup for the other, at least for simple protein substrates [22,23]. Overlaps with other proteostasis network Branches enable integration with the ALP in analogous ways. For example, many transcription factors in the PN regulation Branch are also regulators of the ALP (Figure 6). Moreover, many chaperones in the cytonuclear proteostasis Branch function in autophagy via chaperone-mediated autophagy [75], chaperone-assisted selective autophagy [81],

or endosomal microautophagy [74]. This sharing of components enables autophagy to interface with the folding systems of the cytosol, and to participate as part of the response to folding stress, even though it is not explicitly regulated by cytosolic stress responses like the heat shock response [82].

Discussion

The ALP is one of the two largest Branches of the proteostasis network, consisting of over 850 components and being smaller than only the UPS. The size of the ALP can be attributed both to the complexity of the physical processes involved in autophagy—membrane nucleation and growth, substrate selection, vesicle trafficking and fusion—and to the extensive regulation of these processes. This regulation happens at all points in the ALP, but mTORC1 [34,37], which regulates autophagy initiation, and PI3KC3 complex 1 [83,84], which regulates the PI(3)P content of the autophagophore, are especially dense nodes for regulatory activity. This level of evolutionary investment into the control systems of the ALP shows how critical it is to tune the level of catabolism to the state of the cell. The importance of such tuning is also demonstrated by the outsized role of the ALP in many diseases, and especially aging-related neurodegenerative diseases, as noted above [11,24,25].

We expect that the list of ALP components described herein will enable a deeper understanding of the ways that failures of the ALP contribute to the pathogenesis of such diseases by enabling those studying them to see the full array of connections between, say, the results obtained from patient-derived multiomic datasets and the ALP. Our survey of ALP components could also reveal new targets for diseases with known links to the ALP, or new roles for the ALP in diseases that are not currently associated with it, enabling the exploration of new therapeutic strategies. These activities will be further enhanced by the placement of our ALP annotation in the context of the rest of the proteostasis network, given the deep interconnection that exists between the ALP and the rest of the systems that maintain proteostasis.

We further expect that our annotation will be especially useful in the search for new therapeutics that correct disease-associated deficiencies (or excesses) in ALP activity. This is especially important in the context of phenotypic screens of ALP activity. Such screens are target-agnostic, and identifying the targets of the screening hits can be challenging. Having our

comprehensive list of candidates organized by their roles in the stages of autophagy could greatly facilitate this often-difficult process [85,86].

Finally, we note that we view the list of annotations presented herein as a first version, which is also shared via the Proteostasis Consortium website (https://proteostasisconsortium.com/pn-annotation/) and through an interactive web resource (https://proteostasisnetwork.streamlit.app/). As new ALP components are discovered and as mechanistic understanding deepens, this framework will evolve, as new evidence is incorporated, existing annotations are refined, and coverage is expanded where appropriate. Please contact us at pnannotation@gmail.com with suggestions for components that we have excluded that should have been included, or vice versa. We aim to provide a durable and adaptable foundation for the study of the ALP as a core Branch of the proteostasis network.

Methods

Our "entity-based" and "domain-based" criteria for inclusion of components in the proteostasis network are as follows. "Entity-based" refers to inclusion of an "entity" (a protein or non-coding RNA) based on experimental evidence in the literature that it functions in proteostasis. "Domain-based" refers to inclusion based on a protein containing a type of structural domain that is strongly associated with proteostasis. To generate the initial lists of ALP components we relied on the existing autophagy-related Gene Ontology annotations [28,64] the KEGG pathway "Autophagy - Animal" [65], and previous compilations of ALP components [29-32] supplemented with recent reviews of either autophagy in general or specific aspects of autophagy [11-18,24,34,54,56-59,61-63,73-75,87-90]. The initial list consisted of ~730 components. We then explored the literature for genes with associations with autophagy that were either newly discovered or not recognized in the sources referenced above. We used the Gene database from NCBI [91] to find papers related to each gene in the genome that had been published as of June 2021. The titles, abstracts, and MeSH terms for these papers were downloaded and scanned for the word fragment "-autopha-" to identify papers that could conceivably report a role for the gene of interest in autophagy. We selected ~420 genes for further evaluation that had four or more related papers in which the fragment "-autopha-" occurred and were not already in our preliminary list. The relevant abstracts (and the manuscripts themselves, as necessary) for each of these genes were

read by members of the Proteostasis Consortium to determine if they should be considered as candidates for inclusion in the ALP. Candidate genes were proposed to the Consortium's proteostasis network annotation subgroup and inclusion and exclusion decisions were made collectively by the subgroup. Approximately 110 components were added to the preliminary list through this process. Since 2021, we have continued to add new components if compelling evidence is presented in the literature for them having a role in autophagy. Each ALP component entry in Table S1 (see the Data Availability Statement at the end of the main text for a link to download Table S1) has a note justifying its inclusion in the ALP and explaining the annotation, as well as links to supporting literature, in the last few columns of the table.

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Disclosure statement

The authors declare no competing interests.

Data availability statement

Supplementary Information can be found on the Proteostasis Consortium website (privately-advertised link): https://sites.northwestern.edu/proteostasis-institute/data-download/. Navigate to the header "Manuscript 2 / Survey of the Human Proteostasis Network: The Autophagy-Lysosome Pathway". This study references Table S1 under that heading. The related survey of the translation, protein folding, and organelle-specific systems also references a single table (Table S1 under the corresponding heading), while the survey of the UPS references seven (Tables S1-S7). The annotation in Table S1 is precisely the same for each study, varying only in

which Branches are additionally displayed for the convenience of the reader. Prior to publication, this information will be supplied through the Dryad data repository.

Supplementary Information

Table S1. Human Proteostasis Network Annotation

The complete annotation of the Human Proteostasis Network is presented in the Main tab. The annotation of the ALP Branch begins at Row 1502 in the Main tab and goes through Row 2870. The ALP Branch is additionally presented by itself in the ALP_only tab. Detailed information about the tabs and fields is given in the README tab.

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Tables

Table 1. Groups, Types and Subtypes within the "autophagosome closure, maturation, and lysosome fusion" Class.

Group	Type	Subtype	Components
Sealing of autophagophore membrane	ESCRT-I complex component	(no Subtype)	10
	ESCRT-III complex component	(no Subtype)	11
	ESCRT-III complex activity modulator	(no Subtype)	4
	Localization of the ESCRT-III complex	(no Subtype)	3
	Specific function in sealing of autophagophore membrane unknown	(no Subtype)	2
Localization of the autophagosome	Movement of autophagosomes along microtubules	ATG8 homolog	1
		Recruitment of dynein-dynactin motor complex	1
		HOPS-BORC complex bridging	4
		Miscellaneous function - movement of autophagosomes along microtubules	2
	Movement of autophagosomes along actin	(no Subtype)	4
	Retrograde transport along axons	(no Subtype)	2
	Recruitment of autophagosome to vicinity of ER	(no Subtype)	2
Regulation of autophagosome membrane composition	ATG8 homolog de-lipidation	Releases ATG8 homologs from maturing autophagosome	4
	PIKFYVE complex component	(no Subtype)	3
	Miscellaneous function - regulation of autophagosome membrane composition	(no Subtype)	3
Class 3 PI3K complex 2, direct	Class 3 PI3K complex 2 component	(no Subtype)	7
	Modulator of class 3 PI3K complex 2 activity	(no Subtype)	6
Autophagosome-endosome docking	Lysosome-endosome SNARE complex	(no Subtype)	3
Autophagosome-lysosome docking	Tethering factor recruitment	(no Subtype)	10
	Lysosome-autophagosome SNARE complex	(no Subtype)	6
	Lysosome-autophagosome SNARE complex regulator	(no Subtype)	7
	BORC complex component	(no Subtype)	8
	HOPS complex component	(no Subtype)	6
	HOPS-BORC interaction mediator	(no Subtype)	4
	Recruitment of HOPS complex to autophagosome	ATG8 homolog	6
		ATG8 homolog phosphorylation	2
		Bridges ATG8 and HOPS complex	1
		Bridges STX17, RUBCNL and HOPS complex	1
		Activator of PLECKHM1	2
	ATG8-LAMP1 interaction modulator	(no Subtype)	1
	ATG8-LAMP2 interaction modulator	(no Subtype)	1
	Lysosome-autophagosome interaction modulator	(no Subtype)	13
Localization of the lysosome	(no Type)	(no Subtype)	1
Ca^{2+} efflux	(no Type)	(no Subtype)	2

Specific function in autophagosome maturation and lysosome fusion unknown	(no Type)	(no Subtype)	12

Figure Legends

Figure 1. The proteostasis network. Cartoon representation of a eukaryotic cell with the nine Branches of the proteostasis network in our taxonomic system, shown in boxed labels. The number of components and entries is indicated for the Branches covered in this manuscript. "Components" correspond to unique genes, whereas "entries" correspond to annotation instances; individual components may be assigned to multiple proteostasis network categories and therefore contribute to more than one entry. Created in part in BioRender.

Figure 2. Branches and Classes of the human proteostasis network. The nine Branches of the proteostasis network are shown in two center columns. A Branch comprises multiple Classes, shown flanking each Branch. Progressive subdivisions of the Classes include Groups, Types, and Subtypes (not shown; see Table S1).

Figure 3. A schematic overview of the ALP. The temporal stages that correspond to Classes in the ALP are highlighted. **Pre-initiation autophagy signaling**: Flux through the ALP is regulated by signaling pathways that are upstream of autophagy initiation. **Autophagophore initiation and elongation**: Once autophagy is initiated, the autophagophore double membrane begins to grow. Initiation sometimes occurs at structures in the ER membrane known as omegasomes, but there are likely many mechanisms by which autophagophores can be nucleated. The growing autophagophore is decorated with PI(3)P (phosphatidylinositol 3-phosphate) and ATG8-PE (ATG8-phosphatidylethanolamine conjugate). **Autophagy substrate selection**: As the autophagophore grows, it encloses whatever components of the cytosol are in its vicinity (in macroautophagy) or components that are recruited via interactions between ATG8 and autophagy receptors (in selective autophagy). **Autophagosome closure, maturation, and lysosome fusion**: When it is large enough, the autophagophore membrane closes to form the autophagosome. As the autophagosome is transported in the cell, its membrane composition is modified in the process of autophagosome maturation, which readies it for fusion with the lysosome to form the autolysosome. **Lysosomal catabolism**: After lysosomal fusion, the autophagosome's interior membrane and the cargo it contains are digested by lysosomal enzymes. **Autophagic lysosome reformation:** After lysosomal catabolism is complete, the components of the autolysosome are recycled to form new lysosomes. Classes not pictured: autophagy gene expression, microautophagy, chaperone-mediated autophagy, and specific function in autophagy unknown.

Figure 4. Illustration of the five-level taxonomic scheme applied to proteostasis network components herein using the cysteine protease ATG4A. Blank boxes correspond to categories within each taxonomic level to which ATG4A does not belong. ATG4A has two annotations (see text), but here we illustrate only ATG4A's annotation in the "autophagophore initiation and elongation" Class.

Figure 5. ALP components with annotations in more than one Class within the ALP Branch. Venn diagram shows all ten ALP Classes. Only the 123 (out of 856) ALP components annotated in more than one ALP Class are shown, listed with their official gene symbols.

Figure 6. ALP components annotated in other proteostasis network Branches. Components with annotations in the ALP and in at least one other proteostasis network Branch are shown, listed with their official gene symbols. Only the 223 (out of 856) ALP components annotated in more than one proteostasis network Branch are shown

Figures

Figure 1. The proteostasis network.

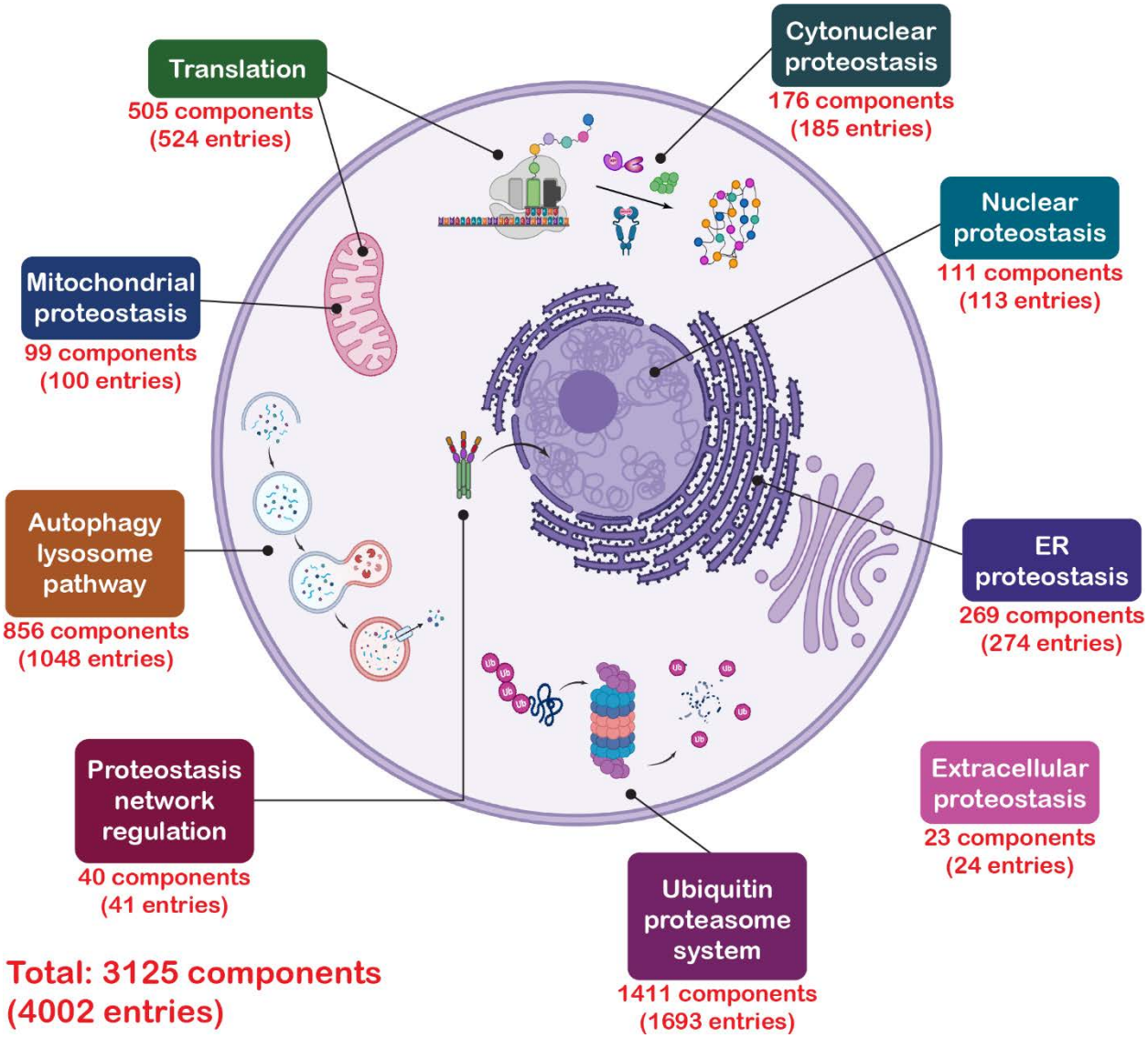

Figure 2. Branches and Classes of the human proteostasis network.

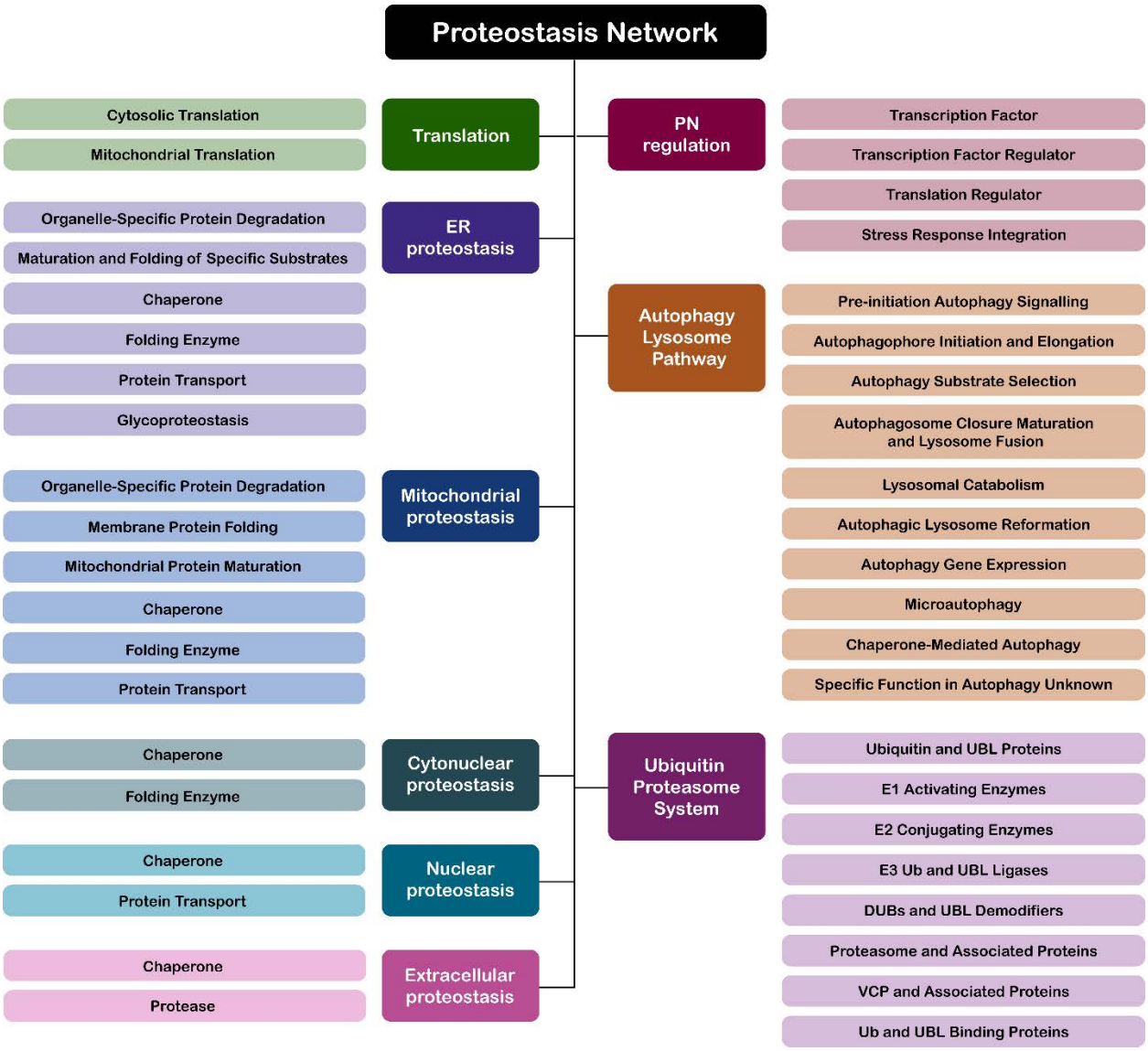

Figure 3. A schematic overview of the ALP.

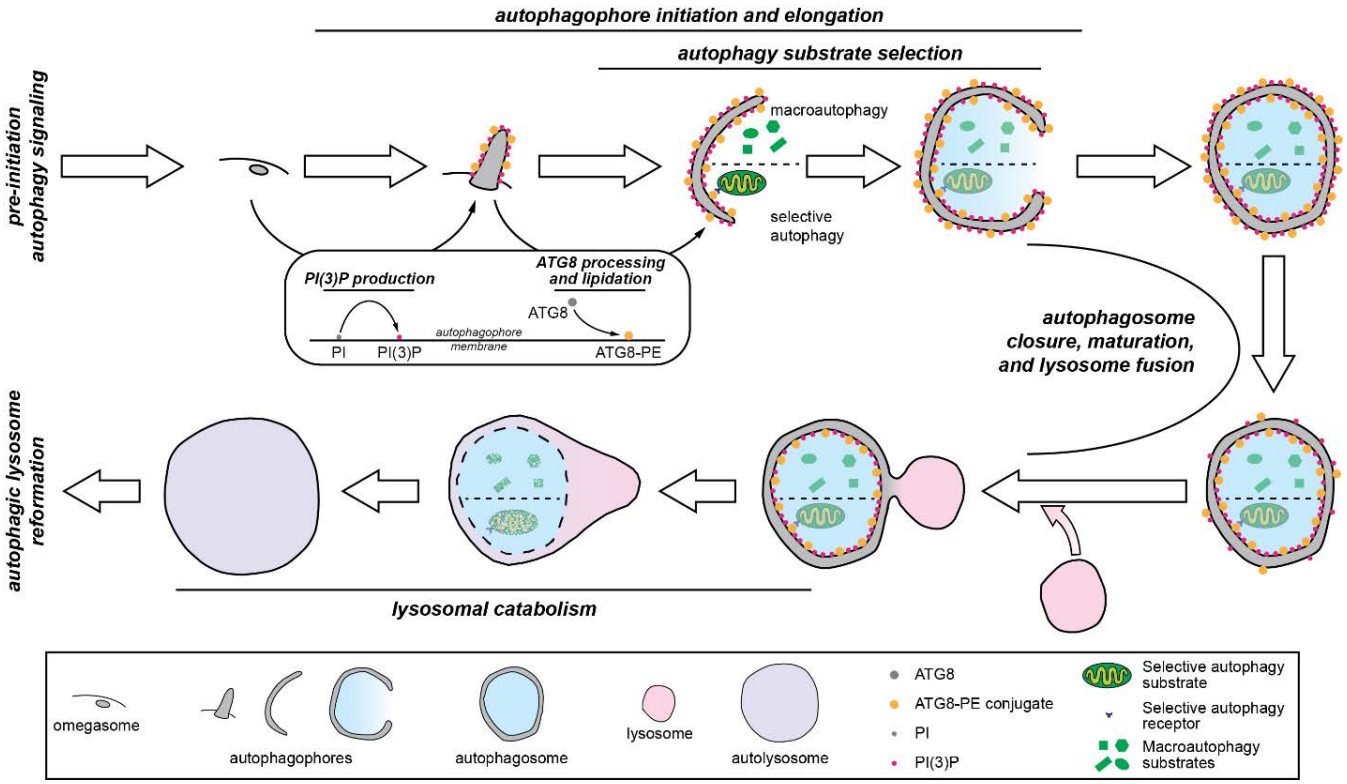

Figure 4. Illustration of the five-level taxonomic scheme applied to proteostasis network components herein using the cysteine protease ATG4A.

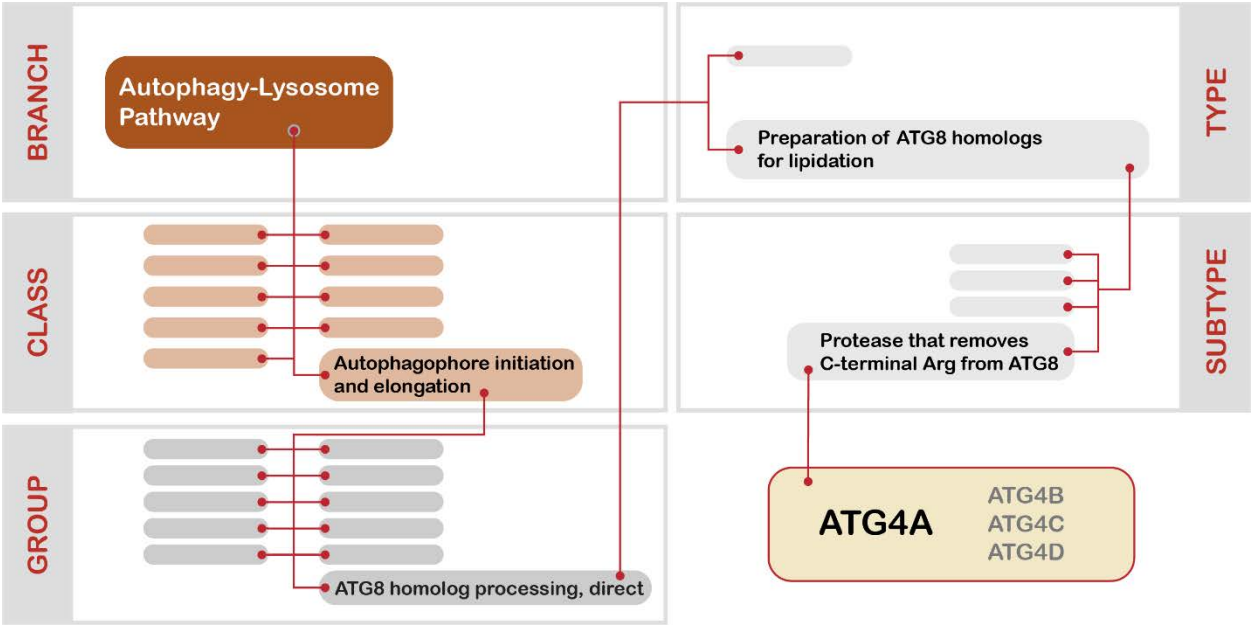

Figure 5. ALP components with annotations in more than one Class within the ALP Branch.

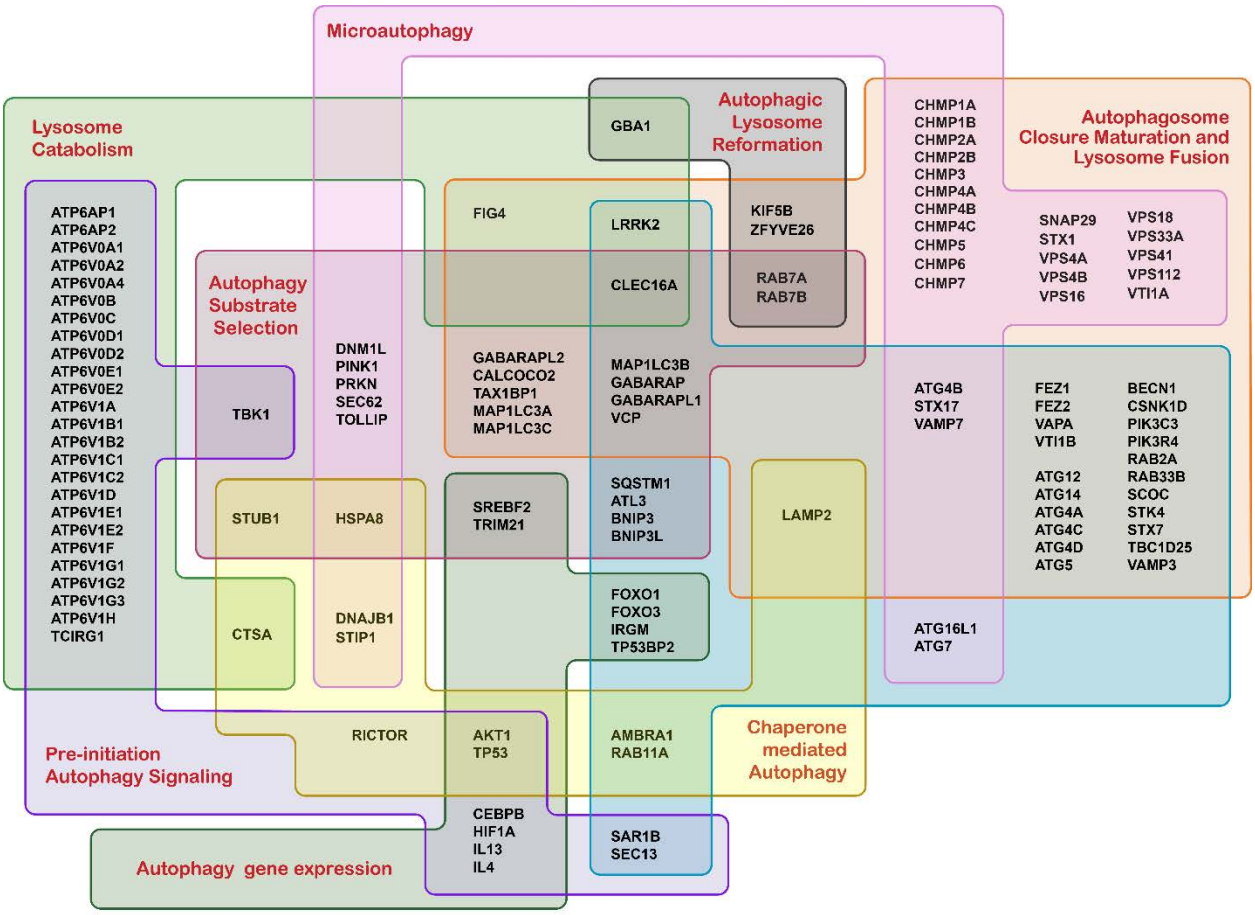
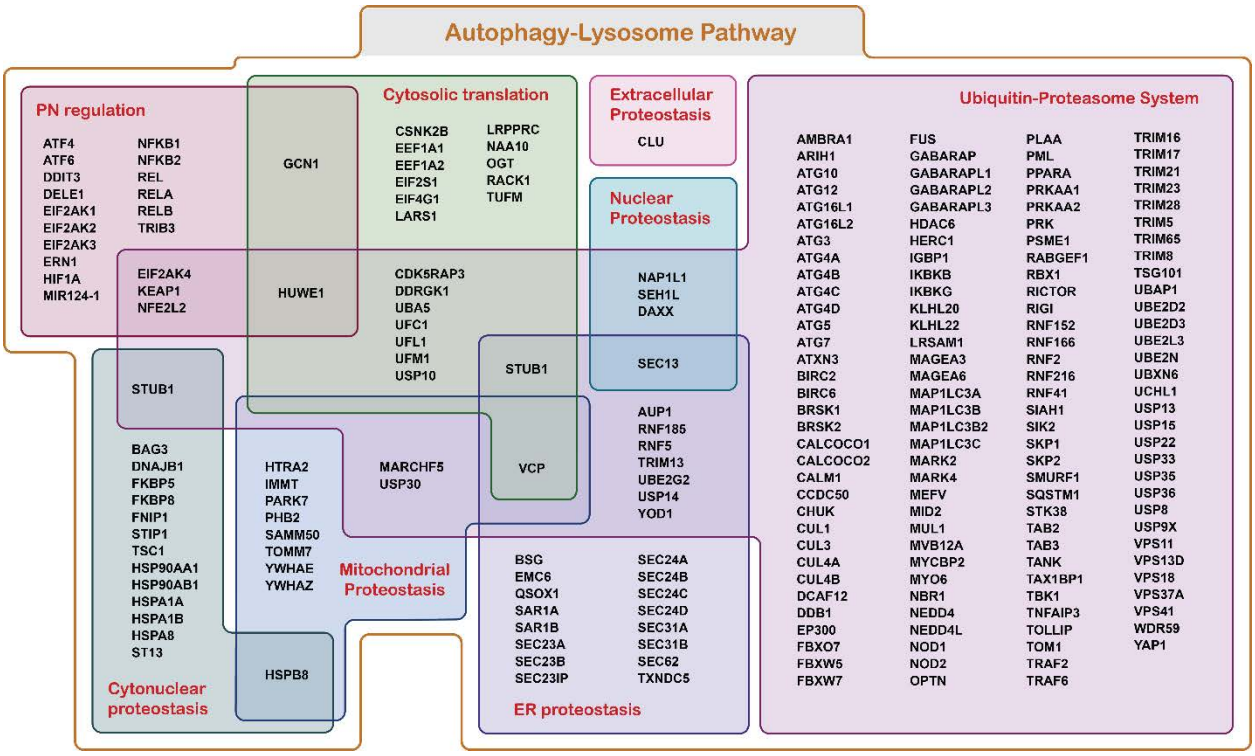
Figure 6. ALP components annotated in other proteostasis network Branches.